# Lesson 64: Barron's Theorem and Approximation Rates

AI/ML Mathematics

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# 1 Motivation and Overview

Lesson 63 established that single-hidden-layer networks with any continuous non-polynomial activation are universal approximators: they can approximate any continuous function on a compact set to arbitrary accuracy. However, the proof was non-constructive and said nothing about *how many neurons are needed*. For generic continuous functions on $[0,1]^d$, the required width grows as $\Omega(\varepsilon^{-d})$—an exponential dependence on dimension that renders the guarantee vacuous in high dimensions.

This raises a fundamental question: *do neural networks offer any advantage over classical methods (polynomials, wavelets, splines) that also suffer the curse of dimensionality?* Barron's 1993 theorem provides a striking affirmative answer: for functions whose Fourier transform has a finite first moment (the *Barron class*), a single-hidden-layer network with $m$ neurons achieves $L^2$ approximation error $O(1/m)$, **independent of the input dimension** $d$. The proof is constructive, based on representing the target function as an expectation over random neurons and approximating this expectation via Monte Carlo sampling. This dimension-free rate explains, in part, why neural networks succeed in high-dimensional problems where classical methods fail.

**Why Barron's Theorem Matters for RL/ML**

- **Value function approximation.** In deep RL, the state space is often high-dimensional ($d \gg 1$). If the optimal value function $V^*$ lies in the Barron class, then Barron's theorem guarantees $O(1/m)$ approximation error with $m$ neurons, independent of $d$—a much stronger guarantee than generic universal approximation.
- **Generalization bounds.** Combining Barron's approximation bound with Rademacher complexity bounds yields sample complexity $O(n^{-1/2})$ for learning Barron-class functions, independent of dimension.
- **Overparameterization.** The integral representation of neural networks (Section 6) connects to mean-field analyses of overparameterized networks and explains why adding more neurons improves approximation without overfitting in certain regimes.
- **Random features.** Barron's Monte Carlo proof inspires the random features method of Rahimi and Recht [2008], where randomly initialized neurons are used as a fixed feature map.

**Prerequisites.** Universal approximation for single-hidden-layer networks (Lesson 63). The Fourier transform on $\mathbb{R}^d$ and the inversion formula (Lesson 3, or any standard reference on Fourier analysis). $L^2$ spaces and basic measure theory (Lessons 1–5). Variance of random variables and Monte Carlo estimation (Lesson 12, Chebyshev's inequality).

**Learning objectives.** After completing this lesson, you will be able to:

- **Define** the Barron norm and the Barron class, and **verify** membership for specific functions (Gaussian, logistic).
- **State and prove** Barron's approximation theorem, including the Monte Carlo sampling argument that yields the $O(1/m)$ rate.
- **Compare** the dimension-free Barron rate with classical approximation rates that depend exponentially on $d$.
- **Derive** the integral representation of neural networks and **explain** its connection to the Barron norm via total variation.

- **Apply** Barron's theorem to bound the approximation error of value function approximation in a high-dimensional RL problem.
- **Explain** how the Barron-class approximation bound combines with estimation bounds to yield dimension-free sample complexity.

## 2 References

- Bach [2024, Chapter 7] — Modern treatment of Barron's theorem in the learning theory framework, with extensions to variation norms.
- Barron [1993] — The original paper proving dimension-free approximation rates for neural networks.
- Cybenko [1989] — Context for Barron's result: the qualitative universal approximation theorem that Barron quantifies.
- Bach [2024, Chapters 4–5] — Background on Rademacher complexity and generalization bounds needed to convert approximation rates into sample complexity.
- Jones [1992] — Independent discovery of a similar approximation bound using a greedy constructive algorithm.
- Bartlett et al. [2017] — Spectrally-normalized bounds for deep networks, extending Barron-type ideas to multiple layers.

## 3 The Barron Class

We define the class of functions for which Barron's theorem provides dimension-free approximation rates. The definition is based on the Fourier transform.

**Definition 3.1** (Fourier Transform)**.** For an integrable function $f\colon \mathbb{R}^d \to \mathbb{R}$ (i.e., $f \in L^1(\mathbb{R}^d)$), the *Fourier transform* of $f$ is

$$\hat{f}(\omega) = \int_{\mathbb{R}^d} f(x)\, e^{-i\omega^\top x}\, dx, \qquad \omega \in \mathbb{R}^d.$$

If additionally $\hat{f} \in L^1(\mathbb{R}^d)$, the *Fourier inversion formula* gives

$$f(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \hat{f}(\omega)\, e^{i\omega^\top x}\, d\omega.$$

**Definition 3.2** (Barron Norm)**.** For a function $f\colon \mathbb{R}^d \to \mathbb{R}$ whose Fourier transform $\hat{f}$ exists, the *Barron norm* (or *Fourier moment*) is

$$C_f \;=\; \int_{\mathbb{R}^d} \|\omega\|_1\, |\hat{f}(\omega)|\, d\omega,$$

where $\|\omega\|_1 = |\omega_1| + \cdots + |\omega_d|$ is the $\ell^1$ norm of the frequency vector.

*Remark* 3.3. Some authors use $\|\omega\|_2$ instead of $\|\omega\|_1$ in the definition. Since $\|\omega\|_1 \le \sqrt{d}\,\|\omega\|_2$ and $\|\omega\|_2 \le \|\omega\|_1$, the two definitions give Barron norms that differ by at most a factor of $\sqrt{d}$. We use $\|\omega\|_1$ to match the convention in Barron [1993].

**Definition 3.4** (Barron Class)**.** The *Barron class* is the set of functions with finite Barron norm:

$$\mathcal{B} = \{f\colon \mathbb{R}^d \to \mathbb{R} : C_f < \infty\}.$$

More generally, for $C > 0$, the *Barron ball* of radius $C$ is $\mathcal{B}_C = \{f \in \mathcal{B} : C_f \le C\}$.

*Remark* 3.5. The condition $C_f < \infty$ can be interpreted as a smoothness requirement. If $f$ is differentiable with gradient $\nabla f$ and $\nabla f \in L^1(\mathbb{R}^d)$, then by the Fourier transform property $\widehat{\partial f/\partial x_j}(\omega) = i\omega_j \hat{f}(\omega)$, we have

$$C_f = \int_{\mathbb{R}^d} \|\omega\|_1 \, |\hat{f}(\omega)| \, d\omega \leq \sum_{j=1}^{d} \int_{\mathbb{R}^d} |\omega_j| \, |\hat{f}(\omega)| \, d\omega.$$

Functions in the Barron class have Fourier transforms that decay fast enough (relative to $\|\omega\|$) to make this integral finite. Roughly speaking, $f \in \mathcal{B}$ if $f$ has "one derivative" in an $L^1$ Fourier sense.

**Example 3.6** (Gaussian Function)**.** Let $f(x) = \exp(-\|x\|_2^2/2)$ for $x \in \mathbb{R}^d$. Its Fourier transform is $\hat{f}(\omega) = (2\pi)^{d/2} \exp(-\|\omega\|_2^2/2)$. Then

$$C_f = (2\pi)^{d/2} \int_{\mathbb{R}^d} \|\omega\|_1 \, e^{-\|\omega\|_2^2/2} \, d\omega.$$

By symmetry and independence of coordinates,

$$C_f = (2\pi)^{d/2} \cdot d \int_{\mathbb{R}^d} |\omega_1| \, e^{-\|\omega\|_2^2/2} \, d\omega = (2\pi)^{d/2} \cdot d \cdot \sqrt{2/\pi} \cdot (2\pi)^{(d-1)/2} = d \cdot 2 \cdot (2\pi)^{d-1}.$$

Since $C_f$ is finite (for each fixed $d$), the Gaussian belongs to $\mathcal{B}$. Note that while $C_f$ grows with $d$, it grows only polynomially ($C_f = O(d)$ up to universal constants after normalization), which is benign compared to the exponential dependence $\varepsilon^{-d}$ in the curse of dimensionality.

**Example 3.7** (Logistic Function)**.** For $d = 1$, the logistic function $f(t) = 1/(1+e^{-t})$ has Fourier transform $\hat{f}(\omega) = \pi/\sinh(\pi\omega) + \pi\delta(\omega)$ (in the distributional sense, accounting for the non-zero limit at $\pm\infty$). The integral $\int |\omega| \, |\hat{f}(\omega)| \, d\omega$ converges because $|\pi/\sinh(\pi\omega)| \sim 2\pi e^{-\pi|\omega|}$ decays exponentially. Thus the logistic function (suitably centered) belongs to $\mathcal{B}$.

**Example 3.8** (Indicator Functions Are Not in the Barron Class)**.** The indicator function $f = \mathbf{1}_{[0,1]^d}$ has Fourier transform $\hat{f}(\omega) = \prod_{j=1}^{d} \frac{1-e^{-i\omega_j}}{i\omega_j}$, so $|\hat{f}(\omega)| \sim C/\|\omega\|_\infty$ for large $\omega$. The integral $\int \|\omega\|_1 \, |\hat{f}(\omega)| \, d\omega$ diverges (it behaves like $\int r^{d-1} \cdot r \cdot r^{-1} \, dr = \int r^{d-1} \, dr$ which diverges). Thus $f \notin \mathcal{B}$. This is expected: indicator functions have discontinuities, and the Barron class captures a form of smoothness.

**The Barron Class Depends on Dimension**

While Barron's *approximation rate* is dimension-free, the Barron *norm* $C_f$ of a given function may depend on $d$. The theorem guarantees $\|f - f_m\|_{L^2}^2 \leq C_f^2/m$; if $C_f$ itself grows exponentially in $d$, the benefit is lost. The dimension-free guarantee is most powerful when $C_f$ is bounded independently of $d$ (or grows at most polynomially).

We now turn to the main result: Barron's theorem, which provides constructive, dimension-free approximation rates for the Barron class.

## 4 Barron's Approximation Theorem

The central idea of Barron's proof is to represent a Barron-class function as an expectation over randomly chosen neurons and then approximate this expectation by an empirical average (Monte Carlo sampling). The $1/m$ rate follows from the variance of the Monte Carlo estimator.

### 4.1 The Fourier Representation as an Expectation

**Lemma 4.1** (Fourier Representation of Barron Functions). *Let $f \in \mathcal{B}$ with $f(0) = 0$ (the centering condition $f(0) = 0$ is for convenience; see Remark 4.2 below). Then $f$ admits the representation*

$$f(x) = \int_{\mathbb{R}^d} (e^{i\omega^\top x} - 1) \frac{\hat{f}(\omega)}{(2\pi)^d}\, d\omega. \tag{1}$$

*Moreover, writing $e^{i\omega^\top x} - 1 = 2i \sin(\omega^\top x/2)\, e^{i\omega^\top x/2}$ and taking real and imaginary parts, $f$ can be expressed as a signed integral over cosine and sine functions:*

$$f(x) = \int_{\mathbb{R}^d} \left[\alpha(\omega)\cos(\omega^\top x) + \beta(\omega)\sin(\omega^\top x)\right] d\omega, \tag{2}$$

*where $\alpha, \beta$ are real-valued functions with $\int_{\mathbb{R}^d} (|\alpha(\omega)| + |\beta(\omega)|)\, \|\omega\|_1\, d\omega \le C_f/(2\pi)^d$.*

*Proof.* By the Fourier inversion formula (Definition 3.1),

$$f(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \hat{f}(\omega)\, e^{i\omega^\top x}\, d\omega.$$

Since $f(0) = 0$, we have $\frac{1}{(2\pi)^d} \int \hat{f}(\omega)\, d\omega = 0$, so

$$f(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \hat{f}(\omega)\, (e^{i\omega^\top x} - 1)\, d\omega,$$

which is (1). Writing $\hat{f}(\omega) = |\hat{f}(\omega)|\, e^{i\phi(\omega)}$ in polar form and expanding $e^{i\omega^\top x} - 1$ into real and imaginary parts gives (2) with

$$\alpha(\omega) = \frac{|\hat{f}(\omega)|}{(2\pi)^d}(\cos\phi(\omega)(\cos(\omega^\top x) - 1) - \sin\phi(\omega)\sin(\omega^\top x)),$$
$$\beta(\omega) = \frac{|\hat{f}(\omega)|}{(2\pi)^d}(\sin\phi(\omega)(\cos(\omega^\top x) - 1) + \cos\phi(\omega)\sin(\omega^\top x)),$$

and the bound on the weighted $L^1$ norm follows from $|\hat{f}(\omega)|/(2\pi)^d$ and the definition of $C_f$. □

*Remark* 4.2. The condition $f(0) = 0$ is not essential. For general $f \in \mathcal{B}$, we write $f(x) = f(0) + g(x)$ where $g(x) = f(x) - f(0)$ satisfies $g(0) = 0$ and $C_g \le C_f + |f(0)|$. The constant $f(0)$ can be absorbed into the output bias of the network.

The key insight is to rewrite the integral in (1) as an *expectation* over a probability distribution on frequencies.

**Proposition 4.3** (Expectation Representation). *Let $f \in \mathcal{B}$ with $f(0) = 0$ and $C_f > 0$. Define the probability measure $\rho$ on $\mathbb{R}^d$ by*

$$d\rho(\omega) = \frac{\|\omega\|_1\, |\hat{f}(\omega)|}{(2\pi)^d\, C_f}\, d\omega,$$

*which is a probability measure since $\int d\rho(\omega) = C_f/((2\pi)^d C_f) \cdot (2\pi)^d = 1$ (by definition of $C_f$, adjusting for the $(2\pi)^d$ factor). Define the function*

$$h(\omega, x) = C_f \cdot \frac{\hat{f}(\omega)}{|\hat{f}(\omega)|\, \|\omega\|_1} \cdot (e^{i\omega^\top x} - 1)$$

*for $\omega$ with $\hat{f}(\omega) \neq 0$. Then*

$$f(x) = \mathbb{E}_{\omega\sim\rho}[h(\omega, x)]. \tag{3}$$

*Moreover, $|h(\omega, x)| \le 2C_f\, \|x\|_\infty$ for all $\omega$ and $x$ (using $|e^{i\omega^\top x} - 1| \le |\omega^\top x| \le \|\omega\|_1 \|x\|_\infty$; and more crudely, $|h(\omega, x)| \le 2C_f$).*

*Proof.* Substituting $d\rho$ into the expectation:

$$\begin{aligned}\mathbb{E}_{\omega\sim\rho}[h(\omega,x)] &= \int_{\mathbb{R}^d} C_f \cdot \frac{\hat{f}(\omega)}{|\hat{f}(\omega)|\,\|\omega\|_1} \cdot (e^{i\omega^\top x}-1) \cdot \frac{\|\omega\|_1\,|\hat{f}(\omega)|}{(2\pi)^d\,C_f}\,d\omega \\ &= \frac{1}{(2\pi)^d}\int_{\mathbb{R}^d} \hat{f}(\omega)\,(e^{i\omega^\top x}-1)\,d\omega \\ &= f(x),\end{aligned}$$

where the last equality uses Lemma 4.1. The bound $|e^{i\theta}-1| \le 2$ gives $|h(\omega,x)| \le 2C_f$. □

## 4.2 Monte Carlo Approximation

Equation (3) expresses $f(x)$ as an expectation. The standard technique for approximating an expectation is Monte Carlo sampling: draw $\omega_1,\dots,\omega_m \overset{\text{iid}}{\sim} \rho$ and set

$$f_m(x) = \frac{1}{m}\sum_{j=1}^{m} h(\omega_j,x).$$

Each term $h(\omega_j,x)$ can be expressed (after separating real and imaginary parts) as a linear combination of $\cos(\omega_j^\top x)$ and $\sin(\omega_j^\top x)$, which is a single neuron with a sinusoidal activation. For sigmoidal activations, we use the fact that sigmoidal functions can approximate sinusoids (or work directly with the activation $\sigma$ and adjust the argument).

The following theorem makes this precise for a general bounded activation.

**Theorem 4.4** (Barron, 1993). *Let $f \in \mathcal{B}$ with Barron norm $C_f < \infty$, and let $B_R = \{x \in \mathbb{R}^d : \|x\|_2 \le R\}$ be a ball of radius $R$. Let $\mu$ be any probability measure supported on $B_R$. Then for every $m \ge 1$, there exist parameters $(a_j, w_j, b_j)_{j=1}^m$ such that the single-hidden-layer network*

$$f_m(x) = a_0 + \sum_{j=1}^{m} a_j\,\sigma(w_j^\top x + b_j)$$

*(with $\sigma$ any bounded sigmoidal activation, and $a_0$ a bias term) satisfies*

$$\int_{B_R} |f(x)-f_m(x)|^2\,d\mu(x) \;\le\; \frac{(2C_fR)^2}{m}. \tag{4}$$

*The bound is independent of the dimension $d$.*

*Proof.* We prove the result for the cosine activation $\sigma(t)=\cos(t)$ (which suffices to establish the rate; the extension to sigmoidal activations follows by the universality of cosines in representing Barron functions, or by a separate argument using sigmoidal functions to approximate cosines with controlled error).

**Step 1: Probabilistic construction.** By Proposition 4.3, there exists a probability distribution $\rho$ on $\mathbb{R}^d$ and a bounded function $h$ such that $f(x) = \mathbb{E}_{\omega\sim\rho}[h(\omega,x)]$ with $|h(\omega,x)| \le 2C_f$ for $x \in B_R$ (using the bound $|e^{i\omega^\top x}-1| \le |\omega^\top x| \le \|\omega\|_1\|x\|_\infty \le \|\omega\|_1 R$, and more carefully $|h(\omega,x)| \le 2C_fR$ after including the $\|\omega\|_1$ normalization). Taking real parts and adjusting, we can write

$$f(x) = \mathbb{E}_{\omega\sim\rho}[\tilde{h}(\omega,x)]$$

where each $\tilde{h}(\omega,\cdot)$ is a single neuron (a cosine function of $\omega^\top x$ plus a phase shift, scaled by a coefficient bounded by $2C_fR$).

**Step 2: Monte Carlo sampling.** Draw $\omega_1, \ldots, \omega_m \overset{\text{iid}}{\sim} \rho$ and define

$$f_m(x) = \frac{1}{m} \sum_{j=1}^{m} \tilde{h}(\omega_j, x).$$

For each fixed $x$, $f_m(x)$ is the sample mean of $m$ i.i.d. random variables, each with mean $f(x)$ and bounded in $[-2C_f R, 2C_f R]$. By the bias-variance decomposition:

$$\mathbb{E}_{\omega_1,\ldots,\omega_m}\left[|f(x) - f_m(x)|^2\right] = \frac{1}{m} \operatorname{Var}_{\omega\sim\rho}[\tilde{h}(\omega, x)] \leq \frac{(2C_f R)^2}{m},$$

since $\operatorname{Var}[Z] \leq \mathbb{E}[Z^2] \leq (\sup |Z|)^2 \leq (2C_f R)^2$.

**Step 3: Integrating over $x$.** Integrating both sides with respect to $\mu$:

$$\mathbb{E}_{\omega_1,\ldots,\omega_m}\left[\int_{B_R} |f(x) - f_m(x)|^2 \, d\mu(x)\right] \leq \frac{(2C_f R)^2}{m}.$$

This means the *expected* $L^2(\mu)$ error of the random approximation $f_m$ is at most $(2C_f R)^2/m$.

**Step 4: Derandomization.** Since the expected error is at most $(2C_f R)^2/m$, there must exist a specific realization $(\omega_1^*, \ldots, \omega_m^*)$ for which

$$\int_{B_R} |f(x) - f_m^*(x)|^2 \, d\mu(x) \leq \frac{(2C_f R)^2}{m}.$$

The corresponding network $f_m^*$ has $m$ neurons with weights $w_j = \omega_j^*$ and output coefficients determined by $\tilde{h}$. This is a valid single-hidden-layer network, establishing (4). □

**Barron's Dimension-Free Approximation Rate**

For any $f$ in the Barron class with norm $C_f$ and any probability measure $\mu$ supported on a ball of radius $R$ in $\mathbb{R}^d$:

$$\inf_{f_m \in \mathcal{F}_m^\sigma} \|f - f_m\|_{L^2(\mu)}^2 \leq \frac{(2C_f R)^2}{m}.$$

The rate $O(1/m)$ is **independent of the dimension** $d$.

**Example 4.5** (Approximating a Gaussian)**.** Let $f(x) = \exp(-\|x\|_2^2/2)$ on $B_R = \{x : \|x\|_2 \leq R\}$. From Example 3.5 of Lesson 63 (or our Example 3.6 above in Section 3), $C_f = O(d)$ (up to constants depending on the normalization). Barron's theorem gives

$$\|f - f_m\|_{L^2(\mu)}^2 \leq \frac{O(d^2 R^2)}{m}.$$

To achieve $\varepsilon$-accuracy, we need $m = O(d^2 R^2/\varepsilon)$ neurons. This is polynomial in $d$, in stark contrast to the $\Omega(\varepsilon^{-d})$ requirement for generic Lipschitz functions.

**Example 4.6** (Approximating a Linear Combination of Sigmoids)**.** Let $f(x) = \sum_{k=1}^{K} c_k \, \sigma(\xi_k^\top x + \beta_k)$ where $\sigma$ is the logistic sigmoid, $\sum_{k=1}^{K} |c_k| \leq C$, and $\max_k \|\xi_k\|_1 \leq W$. Then $f$ belongs to the Barron class with $C_f \leq C \cdot W$ (since each sigmoid has Barron norm $O(\|\xi_k\|_1)$ and the Barron norm satisfies the triangle inequality). Barron's theorem gives $\|f - f_m\|_{L^2}^2 \leq (2CWR)^2/m$. Note that the bound depends on the total weight $C$ and the norm of the weight vectors $W$, not on the original number of neurons $K$.

*Remark* 4.7. Unlike Cybenko's proof (which used the Hahn–Banach theorem and was non-constructive), Barron's proof is *constructive* in principle: one can draw random frequencies $\omega_j \sim \rho$ and obtain a network achieving the $O(1/m)$ rate. This is the idea behind the *random features* method of Rahimi and Recht [2008], which uses randomly initialized neurons as a fixed feature map and then optimizes only the output weights $a_j$ via linear regression.

**Random Features in RL**

In large-scale RL, random features provide a computationally efficient alternative to full neural network training. One initializes $m$ neurons with random weights $w_j$ (drawn from a distribution related to $\rho$) and then learns only the output weights $a_j$ by solving a linear regression problem:

$$\hat{V}(s) = \sum_{j=1}^{m} a_j\, \sigma(w_j^\top s + b_j), \qquad \hat{a} = \arg\min_a \sum_{i=1}^{n} \big(V_i - \sum_{j=1}^{m} a_j\, \sigma(w_j^\top s_i + b_j)\big)^2.$$

Since the problem is linear in $a$, it is convex and can be solved in closed form. Barron's theorem guarantees that the approximation error from using $m$ random features is $O(1/m)$ when the target function is in the Barron class.

The next section explores the broader implications of Barron's result and compares it to classical approximation theory.

# 5 Implications and Extensions

## 5.1 Comparison with Classical Approximation Theory

Classical nonparametric methods (polynomials, splines, wavelets) achieve approximation rates that depend on the smoothness of the target function and the dimension of the input space. For a function with $s$ derivatives on $[0,1]^d$, the optimal approximation rate using $m$ parameters is $O(m^{-2s/d})$.

**Proposition 5.1** (Classical Approximation Rates)**.** *Let $\mathcal{W}^{s,2}([0,1]^d)$ denote the Sobolev space of functions with $s$ square-integrable derivatives. For any linear approximation method using $m$ parameters:*

$$\inf_{\dim(\textit{approximation space})=m} \sup_{f\in\mathcal{W}^{s,2}([0,1]^d)} \|f - \hat{f}_m\|_{L^2} \;=\; \Theta(m^{-s/d}).$$

*Proof.* This is a classical result in approximation theory; see Bach [2024, Chapter 7] for a discussion in the neural network context. The lower bound follows from information-theoretic arguments (Kolmogorov $n$-widths), and the upper bound is achieved by tensor-product polynomial or wavelet expansions. □

**Example 5.2** (Dimension Dependence: Sobolev vs. Barron)**.** Consider approximating a function with one derivative ($s = 1$) in $d = 100$ dimensions.

- **Classical (Sobolev):** Rate $O(m^{-1/100})$. To achieve $\varepsilon = 0.01$, need $m = (1/0.01)^{100} = 10^{200}$ parameters.
- **Neural network (Barron):** Rate $O(C_f^2/m)$. If $C_f$ is moderate (say $C_f = 10$), need $m = (2\cdot 10\cdot R)^2/0.01 = O(10^5)$ neurons (for $R = 1$).

The improvement from $10^{200}$ to $10^5$ illustrates the dramatic advantage of neural networks for Barron-class functions in high dimensions.

*Remark* 5.3. The classical lower bound in Proposition 5.1 applies to *linear* approximation methods (where the approximation space is fixed independently of $f$). Neural networks use *nonlinear* approximation: the "basis functions" $\sigma(w_j^\top x + b_j)$ depend on the target $f$ through the learned weights. This nonlinearity is key to escaping the curse of dimensionality. The Barron class is precisely a class where nonlinear (neural network) approximation is provably better than linear approximation.

### 5.2 Combining Approximation and Estimation

In statistical learning, the total excess risk decomposes as

$$R(\hat{f}_n) - R(f^*) = \underbrace{\big(R(\hat{f}_n) - \inf_{f\in\mathcal{F}_m} R(f)\big)}_{\text{estimation error}} + \underbrace{\big(\inf_{f\in\mathcal{F}_m} R(f) - R(f^*)\big)}_{\text{approximation error}}, \tag{5}$$

where $\hat{f}_n$ is the empirical risk minimizer over $\mathcal{F}_m$ with $n$ samples, $f^*$ is the Bayes-optimal predictor, and $R(f) = \mathbb{E}[\ell(f(X), Y)]$.

**Proposition 5.4** (Sample Complexity for the Barron Class)**.** *Let $f^* \in \mathcal{B}_C$ (Barron class with norm $\leq C$), and let $\hat{f}_n$ be the empirical risk minimizer over $\mathcal{F}_m^\sigma$ using $n$ i.i.d. samples with bounded loss $\ell \in [0,1]$. If we choose $m = n^{1/2}$, then*

$$\mathbb{E}[R(\hat{f}_n)] - R(f^*) \;\leq\; \frac{(2CR)^2}{n^{1/2}} + \frac{c\,n^{1/2}}{n^{1/2}} \;=\; O(n^{-1/2}),$$

*where the first term is the approximation error (Barron's theorem with $m = n^{1/2}$) and the second term is the estimation error (Rademacher complexity of $\mathcal{F}_m$ scales as $O(\sqrt{m/n})$, giving $O(\sqrt{m}/\sqrt{n}) = O(n^{-1/4})$ for the estimation term).*

*More precisely, with careful balancing of $m$ and $n$, the excess risk is $O(C^2R^2/\sqrt{n})$,* ***independent of*** *$d$.*

*Proof sketch.* The approximation error is $O(C^2R^2/m)$ by Theorem 4.4. For the estimation error, the Rademacher complexity of $\mathcal{F}_m^\sigma$ with weights bounded by $B$ satisfies $\mathrm{Rad}_n(\mathcal{F}_m) \leq O(B\sqrt{m\log m/n})$ (by standard norm-based bounds, e.g., Bartlett et al. [2017]). With $B = O(C)$ and balancing the two error terms by setting $m \propto \sqrt{n}$ gives the stated $O(n^{-1/2})$ rate. □

**Dimension-Free Learning Rate**

For target functions in the Barron class with norm $C_f \leq C$ on a ball of radius $R$, neural networks achieve excess risk $O(C^2R^2/\sqrt{n})$ from $n$ samples, **independent of the input dimension** $d$. This stands in contrast to classical nonparametric methods, which achieve $O(n^{-2s/(2s+d)})$—a rate that degrades exponentially with $d$.

### 5.3 Extension: Variation Norm and Path Norm

The Barron norm defined via the Fourier transform is one way to measure the complexity of a function relative to neural networks. An alternative, more general framework uses the *variation norm*.

**Definition 5.5** (Variation Norm)**.** For a function $f$ that can be represented as

$$f(x) = a_0 + \int_{\mathbb{R}^d\times\mathbb{R}} a\,\sigma(w^\top x + b)\,d\nu(a,w,b)$$

for some signed measure $\nu$ on $\mathbb{R}\times\mathbb{R}^d\times\mathbb{R}$, the *variation norm* of $f$ is

$$\|f\|_{\mathcal{V}} = \inf_\nu |\nu|(\mathbb{R}\times\mathbb{R}^d\times\mathbb{R}),$$

where $|\nu|$ is the total variation of $\nu$ and the infimum is over all representations.

*Remark* 5.6. The variation norm generalizes the Barron norm: for smooth activations, $\|f\|_{\mathcal{V}} \leq C\cdot C_f$ for a universal constant $C$ [Bach, 2024, Chapter 7]. The variation norm applies to any activation (not just those with nice Fourier properties) and connects naturally to the integral representation studied in Section 6.

We now develop the integral representation perspective, which provides the natural framework for understanding the Barron norm and its generalizations.

## 6 The Integral Representation Perspective

The Monte Carlo proof of Barron's theorem suggests viewing a neural network not as a finite sum of neurons but as an *integral* over a continuum of neurons, parameterized by a measure on weight space.

**Definition 6.1** (Integral Representation of Neural Networks)**.** Let $\Theta = \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}$ denote the parameter space (output weight, input weights, bias). An *integral neural network* is a function of the form

$$f(x) = a_0 + \int_\Theta a\,\sigma(w^\top x + b)\,d\nu(a, w, b), \tag{6}$$

where $\nu$ is a signed Borel measure on $\Theta$ with finite total variation $|\nu|(\Theta) < \infty$, and $a_0 \in \mathbb{R}$ is a bias term.

*Remark* 6.2. A standard network with $m$ neurons corresponds to the *discrete* measure

$$\nu = \sum_{j=1}^{m} \delta_{(a_j, w_j, b_j)},$$

where $\delta_{(a_j,w_j,b_j)}$ is the Dirac mass at the $j$-th neuron's parameters. In this case, (6) reduces to

$$f(x) = a_0 + \sum_{j=1}^{m} a_j\,\sigma(w_j^\top x + b_j),$$

and $|\nu|(\Theta) = \sum_{j=1}^m |a_j| \cdot 1 = \sum_{j=1}^m |a_j|$ (here $a_j$ is already factored into the Dirac mass, but if we write the measure as $\nu = \sum \delta_{(1,w_j,b_j)}$ with the $a_j$ as the "mass" at each point, then $|\nu|(\Theta)$ corresponds to $\sum |a_j|$). More precisely, if we parameterize as $\nu = \sum_{j=1}^m a_j\,\delta_{(w_j,b_j)}$ (viewing $\nu$ as a measure on $\mathbb{R}^d \times \mathbb{R}$), then $|\nu|(\mathbb{R}^d \times \mathbb{R}) = \sum_{j=1}^m |a_j|$.

**Proposition 6.3** (Barron Norm as Total Variation)**.** *The Barron norm of $f$ (Definition 3.2) satisfies*

$$C_f \;=\; \inf\Big\{|\nu|(\Theta) : f(x) = a_0 + \int_\Theta a\,e^{iw^\top x}\,d\nu(a, w, b)\ \ a.e.\Big\},$$

*i.e., $C_f$ is the infimum of the total variation over all integral representations of $f$ using complex exponential "neurons." For sigmoidal activations, the variation norm (Definition 5.5) provides the analogous characterization.*

*Proof.* The Fourier inversion formula gives $f(x) = (2\pi)^{-d} \int \hat{f}(\omega) e^{i\omega^\top x} d\omega$. We can write this as an integral over the signed measure $d\nu(\omega) = (2\pi)^{-d} \hat{f}(\omega)\,d\omega$, and $|\nu|(\mathbb{R}^d) = (2\pi)^{-d} \int |\hat{f}(\omega)|\,d\omega$. The first-moment weighting $\|\omega\|_1 |\hat{f}(\omega)|$ in the Barron norm arises because Barron's construction separates the constant term ($f(0)$) and uses the factor $e^{i\omega^\top x} - 1$, which satisfies $|e^{i\omega^\top x} - 1| \le \|\omega\|_1 \|x\|_\infty$; the $\|\omega\|_1$ weighting controls the Lipschitz constant of each neuron. The equivalence between the Fourier definition and the total-variation definition (up to constants depending on $R$) is established in Bach [2024, Chapter 7]. □

**Example 6.4** (Integral Representation of a Gaussian)**.** For $f(x) = \exp(-\|x\|_2^2/2)$ on $\mathbb{R}^d$, the integral representation is

$$f(x) = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{i\omega^\top x}\, e^{-\|\omega\|_2^2/2}\,d\omega,$$

which is the Fourier inversion formula. Here $\nu$ is the (signed, in fact positive) measure with density $(2\pi)^{-d/2} e^{-\|\omega\|^2/2}$, and $|\nu|(\mathbb{R}^d) = 1$. The Barron norm is $C_f = (2\pi)^{-d/2} \int \|\omega\|_1 e^{-\|\omega\|^2/2} d\omega = O(d)$, confirming Example 3.6.

**Example 6.5** (Integral Representation of a Two-Layer Network)**.** A standard two-layer network $f(x) = \sum_{j=1}^m a_j \sigma(w_j^\top x + b_j)$ has the integral representation

$$f(x) = \int_{\mathbb{R}^d \times \mathbb{R}} \sigma(w^\top x + b)\, d\nu(w, b), \qquad \nu = \sum_{j=1}^m a_j\, \delta_{(w_j, b_j)}.$$

The total variation is $|\nu| = \sum_{j=1}^m |a_j|$, i.e., the $\ell^1$ norm of the output weights. This connects the Barron/variation norm to the common practice of $\ell^1$ regularization on the output layer.

**Mean-Field Limits and Overparameterization**

The integral representation provides the mathematical foundation for *mean-field* analyses of neural networks:

- As $m \to \infty$, the discrete measure $\nu_m = \frac{1}{m} \sum_{j=1}^m \delta_{(a_j, w_j, b_j)}$ converges (in an appropriate sense) to a continuous measure $\nu$, and the finite network converges to the integral representation.
- Training the network via gradient descent corresponds to a gradient flow on the space of measures. In the mean-field limit, this becomes a partial differential equation (Wasserstein gradient flow) describing the evolution of $\nu$.
- The global convergence results for overparameterized networks (e.g., in the NTK regime) can be understood through this lens: with enough neurons, the discrete measure $\nu_m$ can approximate any continuous $\nu$ with bounded total variation.

These connections are explored further in Chizat and Bach [2018] and Mei et al. [2018].

**Barron's Bound Is Not Tight in General**

The $O(1/m)$ rate in Barron's theorem is an upper bound. For specific functions, much faster rates may be achievable (e.g., if the target is itself a finite neural network, $m$ neurons suffice exactly). Conversely, the $O(1/m)$ rate is known to be optimal (up to constants) over the entire Barron class: there exist Barron-class functions for which no $m$-neuron network achieves error better than $\Omega(1/m)$. The constant $(2C_f R)^2$ in the bound can potentially be improved with more refined analyses.

## 7 Exercises

**Exercise 7.1** (Computing Barron Norms)**.** (*)
Compute the Barron norm $C_f$ (or bound it) for the following functions on $\mathbb{R}$:

(a) $f(t) = e^{-|t|}$ (the Laplace density, unnormalized).

(b) $f(t) = \frac{1}{1+t^2}$ (the Cauchy density, unnormalized).

(c) $f(t) = \sin(t) \cdot e^{-t^2}$.

In each case, determine whether $f$ belongs to the Barron class.

**Exercise 7.2** (Dimension Dependence of the Barron Norm)**.** (*)
Let $f(x) = \prod_{j=1}^d g(x_j)$ where $g\colon \mathbb{R} \to \mathbb{R}$ is a one-dimensional function with Barron norm $C_g < \infty$. Show that the Barron norm of $f$ satisfies $C_f \le d \cdot C_g \cdot (\int |\hat{g}(\omega)|\, d\omega)^{d-1}$. Discuss: for which functions $g$ does $C_f$ remain polynomial in $d$?

**Exercise 7.3** (Optimality of the $1/m$ Rate)**.** (**)
Show that the $O(1/m)$ rate in Barron's theorem is optimal over the Barron class. Specifically:

(a) Let $f(x) = \mathbb{E}_{\omega\sim\rho}[\cos(\omega^\top x)]$ where $\rho$ is a probability measure on $\mathbb{R}^d$ with bounded support. Show that $f \in \mathcal{B}$ and compute $C_f$ in terms of $\rho$.

(b) Show that any approximation of $f$ by $m$ cosine neurons $f_m(x) = \sum_{j=1}^m a_j \cos(\omega_j^\top x + \phi_j)$ satisfies $\mathbb{E}_x[|f(x) - f_m(x)|^2] \geq c \cdot C_f^2/m$ for a constant $c > 0$.

*Hint:* For part (b), use the fact that the variance of the Monte Carlo estimator achieves the Cramer–Rao lower bound.

**Exercise 7.4** (Value Function in the Barron Class)**.** (**)
Consider an MDP with state space $S = \{x \in \mathbb{R}^d : \|x\|_2 \leq 1\}$, discount factor $\gamma = 0.99$, and bounded reward $|r(s,a)| \leq 1$.

(a) Show that $\|V^*\|_\infty \leq 1/(1-\gamma) = 100$.

(b) Suppose the transition kernel $P(s'|s,a)$ is a Gaussian with mean $g(s,a)$ (a smooth function) and covariance $\sigma^2 I$. Argue (informally) that $V^*$ is "smooth" and plausibly belongs to the Barron class.

(c) Using Barron's theorem, bound the number of neurons $m$ needed to approximate $V^*$ within $\varepsilon$ in $L^2$ (in terms of $C_{V^*}$, $d$, and $\varepsilon$). Compare with the generic $\Omega(\varepsilon^{-d})$ bound.

**Exercise 7.5** (Random Features Implementation)**.** (**)
Let $f^*(x) = \sin(\|x\|_2^2)$ on $B_1 = \{x \in \mathbb{R}^d : \|x\|_2 \leq 1\}$, and let $\mu$ be the uniform distribution on $B_1$.

(a) Describe a random features approximation: draw $\omega_1, \ldots, \omega_m$ from a suitable distribution $\rho$ on $\mathbb{R}^d$ and define $\hat{f}_m(x) = \sum_{j=1}^m a_j \cos(\omega_j^\top x + \phi_j)$. How should $\rho$ be chosen?

(b) The output weights $a_j$ are found by least-squares regression on $n$ samples. Write the optimization problem and explain why it is convex.

(c) Using the approximation-estimation decomposition (5), bound the total error as a function of $m$ and $n$. How should $m$ be chosen as a function of $n$?

**Exercise 7.6** (Total Variation and $\ell^1$ Regularization)**.** (***)
Consider training a single-hidden-layer network with $m$ neurons using $\ell^1$-regularized empirical risk minimization:

$$\min_{a_0, a_j, w_j, b_j} \frac{1}{n} \sum_{i=1}^n (y_i - f_m(x_i))^2 + \lambda \sum_{j=1}^m |a_j|,$$

where $f_m(x) = a_0 + \sum_{j=1}^m a_j \sigma(w_j^\top x + b_j)$.

(a) Show that the regularizer $\sum_{j=1}^m |a_j|$ equals the total variation $|\nu_m|$ of the discrete measure $\nu_m = \sum_{j=1}^m a_j \delta_{(w_j, b_j)}$.

(b) Explain how this regularizer controls the variation norm and hence (by Proposition 6.3) relates to the Barron norm of the learned function.

(c) Argue that the $\ell^1$ regularizer promotes sparsity in the output weights and thus favors networks that are well-approximated by fewer neurons.

## 8 Summary

- The **Barron norm** of a function $f$ is $C_f = \int \|\omega\|_1 \, |\hat{f}(\omega)| \, d\omega$, measuring how fast the Fourier transform decays. The **Barron class** $\mathcal{B} = \{f : C_f < \infty\}$ captures functions with a quantifiable degree of smoothness in the Fourier domain.
- **Barron's theorem:** For any $f \in \mathcal{B}$ and any probability measure $\mu$ on a ball of radius $R$ in $\mathbb{R}^d$, there exists a single-hidden-layer network with $m$ neurons achieving $\|f - f_m\|^2_{L^2(\mu)} \leq (2C_f R)^2/m$.
- The approximation rate $O(1/m)$ is **independent of the dimension** $d$, in contrast to the classical $O(m^{-2s/d})$ rate for Sobolev-class functions. This explains why neural networks can succeed in high-dimensional problems.
- The proof is **constructive**: represent $f$ as an expectation over random neurons (via the Fourier inversion formula), then approximate the expectation by Monte Carlo sampling with $m$ samples. The $1/m$ rate follows from the variance of the estimator.
- Combining Barron's approximation bound with Rademacher complexity bounds yields a **dimension-free sample complexity** of $O(n^{-1/2})$ for learning Barron-class functions.
- The **integral representation** $f(x) = \int a \, \sigma(w^\top x + b) \, d\nu(a, w, b)$ views a neural network as parameterized by a measure $\nu$ on weight space. The Barron norm corresponds to the total variation of the optimal measure $\nu$.
- A finite network with $m$ neurons corresponds to a discrete measure $\nu = \sum_{j=1}^m a_j \delta_{(w_j, b_j)}$, with total variation $\sum |a_j|$. This connects the Barron norm to $\ell^1$ regularization of the output weights.
- The integral representation connects to **mean-field theory** of overparameterized networks: as $m \to \infty$, the discrete measure converges to a continuous measure, and training dynamics become a gradient flow in the space of measures.

## 9 Further Reading

- **Lesson 65** addresses the optimization question for training neural networks: SGD convergence, adaptive methods, and the loss landscape geometry.
- **Lesson 66** (RNN) and **Lesson 67** (LSTM) study recurrent architectures; the approximation theory developed here applies to feedforward networks but similar questions arise for sequence models.
- Barron [1993] is the original paper; it also proves a lower bound showing that the $O(1/m)$ rate is optimal for the Barron class, and discusses the connection to information theory and minimum description length.
- Jones [1992] independently proved a similar result using a greedy constructive algorithm (analogous to matching pursuit) rather than Monte Carlo sampling.
- Bartlett et al. [2017] extends Barron-type norm-based bounds to deep networks, using spectral norms of weight matrices to control the Rademacher complexity. This provides dimension-free generalization bounds for deep networks, not just shallow ones.
- Chizat and Bach [2018] and Mei et al. [2018] develop the mean-field theory of two-layer networks, rigorously connecting the integral representation to the training dynamics of overparameterized networks.

- **Historical note.** Barron's 1993 paper appeared four years after Cybenko's and Hornik's qualitative results, and it was the first to give a quantitative rate that explained the empirical success of neural networks in high dimensions. The Monte Carlo proof technique was later rediscovered and popularized by Rahimi and Recht [2008] as "random features," becoming a practical tool in large-scale kernel learning.

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